

Feature Engineering & Data Preprocessing on Titanic Dataset

Introduction

Data preprocessing and feature engineering are important steps in the machine learning pipeline. Raw datasets often contain missing values, irrelevant features, and categorical variables that cannot be directly used by machine learning algorithms. This project demonstrates how preprocessing techniques can be applied to the Titanic dataset to transform it into a clean and model-ready dataset.

Objective

The objective of this project is to perform feature engineering and data preprocessing on the Titanic dataset by handling missing values, creating new features, encoding categorical variables, and scaling numerical features.

Dataset Description

The Titanic dataset contains information about passengers aboard the Titanic, including demographic details, ticket information, family relationships, and survival status. The target variable is **Survived**, which indicates whether a passenger survived the disaster.

Preprocessing Techniques Applied

1. Missing Value Handling

Missing values were identified and treated as follows:

- Missing values in the **Age** column were replaced with the median age.
- Missing values in the **Embarked** column were replaced with the most frequent value (mode).
- Missing values in the **Fare** column were replaced with the median fare value.

This ensured that the dataset contained no missing values before model training.

2. Feature Selection

Some columns were removed because they were either unique identifiers or contained excessive missing values and were not expected to contribute significantly to prediction performance.

Removed columns:

- PassengerId
- Name

- Ticket
- Cabin

3. Feature Engineering

New features were created to provide additional useful information to machine learning models.

FamilySize

A new feature named **FamilySize** was created using:

$$\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$$

This feature represents the total number of family members traveling with a passenger.

IsAlone

A new feature named **IsAlone** was created.

- Value 1: Passenger traveled alone.
- Value 0: Passenger traveled with family.

Title

Passenger titles such as Mr, Mrs, Miss, and Master were extracted from the Name column and used as a new feature because titles can provide useful demographic information.

4. Encoding Categorical Variables

Machine learning algorithms require numerical inputs. Therefore:

- Label Encoding was applied to the **Sex** column.
- One-Hot Encoding was applied to **Embarked** and **Title** columns.

This converted categorical data into numerical format.

5. Feature Scaling

Feature scaling was performed to ensure numerical variables were on a comparable scale.

- StandardScaler was applied to Age and Fare.
- MinMaxScaler was applied to FamilySize.

Scaling improves model performance and helps many machine learning algorithms converge more effectively.

Results

After preprocessing:

- All missing values were removed.
- Categorical variables were successfully converted into numerical format.
- Numerical features were scaled.
- New informative features were created.
- The dataset became suitable for machine learning model training.

Features Created

1. FamilySize
2. IsAlone
3. Title

Benefits of Preprocessing

- Improved data quality
- Reduced inconsistencies
- Better representation of passenger information
- Enhanced machine learning readiness
- Potential improvement in predictive performance

Conclusion

This project successfully transformed the raw Titanic dataset into a clean and machine-learning-ready dataset using feature engineering and preprocessing techniques. The preprocessing pipeline included missing value treatment, feature creation, encoding, scaling, and feature selection. The resulting dataset is suitable for use in classification models and serves as a strong foundation for further machine learning tasks.